

33333333SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA0501

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)

Assessment Tool Weightage: 30%

Dr

QUESTIONS			Total
Marks: 50			
Q.No	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by	BL1 – Remember	5

	item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar’s Office Management System A university registrar’s office maintains data about the following entities: - courses, including number, title, credits, syllabus, and prerequisites; - course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; - students, including student-id, name, and program;	BL2 – Understand	5

	d. instructors, including identification number, name, department, and title. e. Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.		
4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. • A department controls a number of PROJECT. Projects have a unique name, number and a single location. • Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The	BL2 – Understand	5

	immediate supervisor for the employee. Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).		
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5
8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

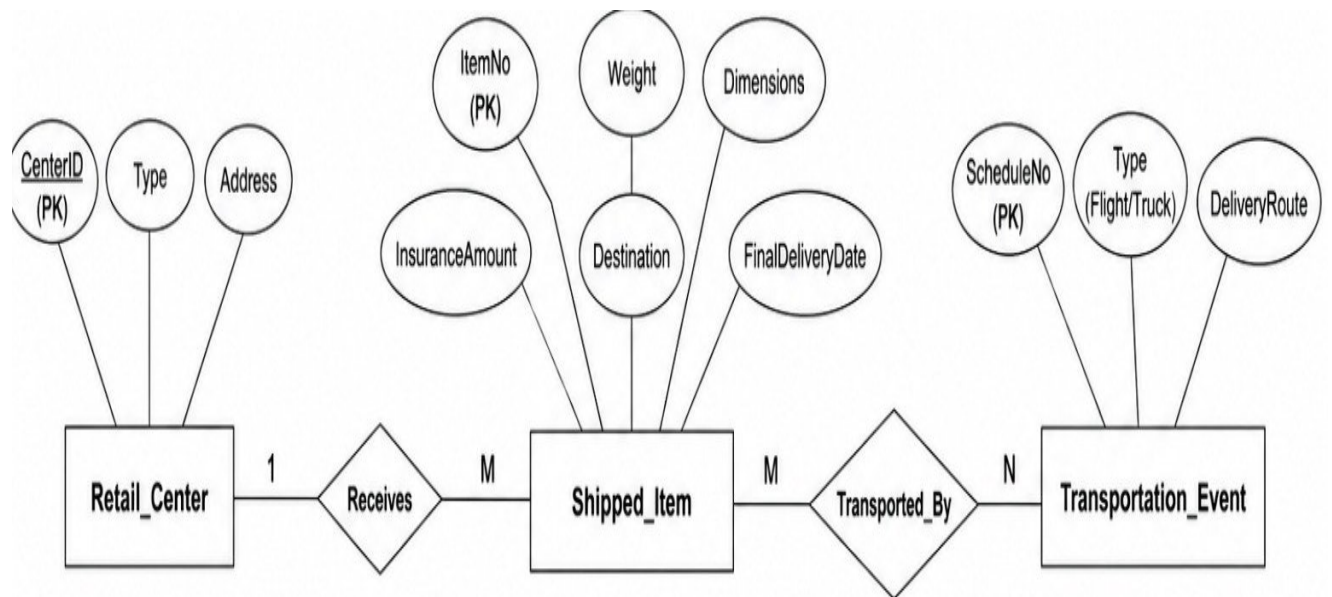
Code of Conduct:

I D. Gnanadeepthi (192411123)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

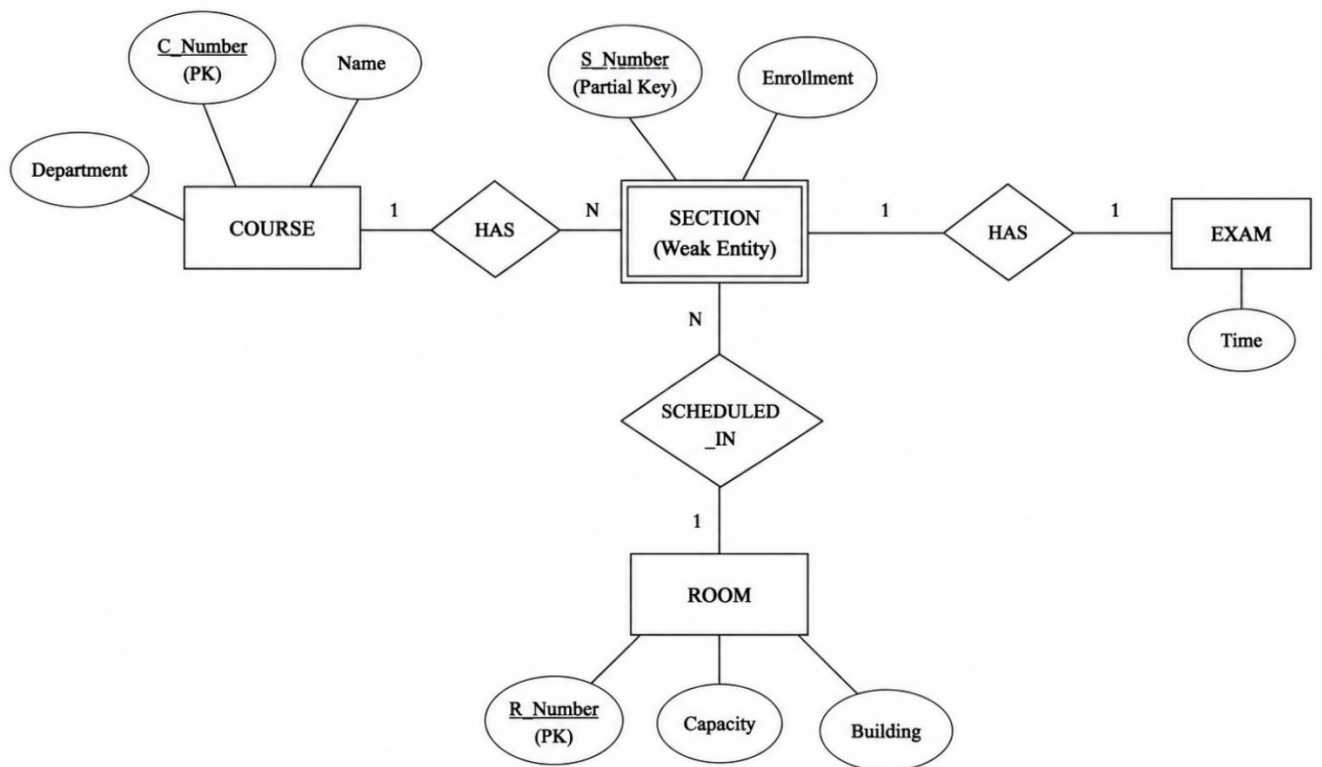
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D. Gnanadeepthi

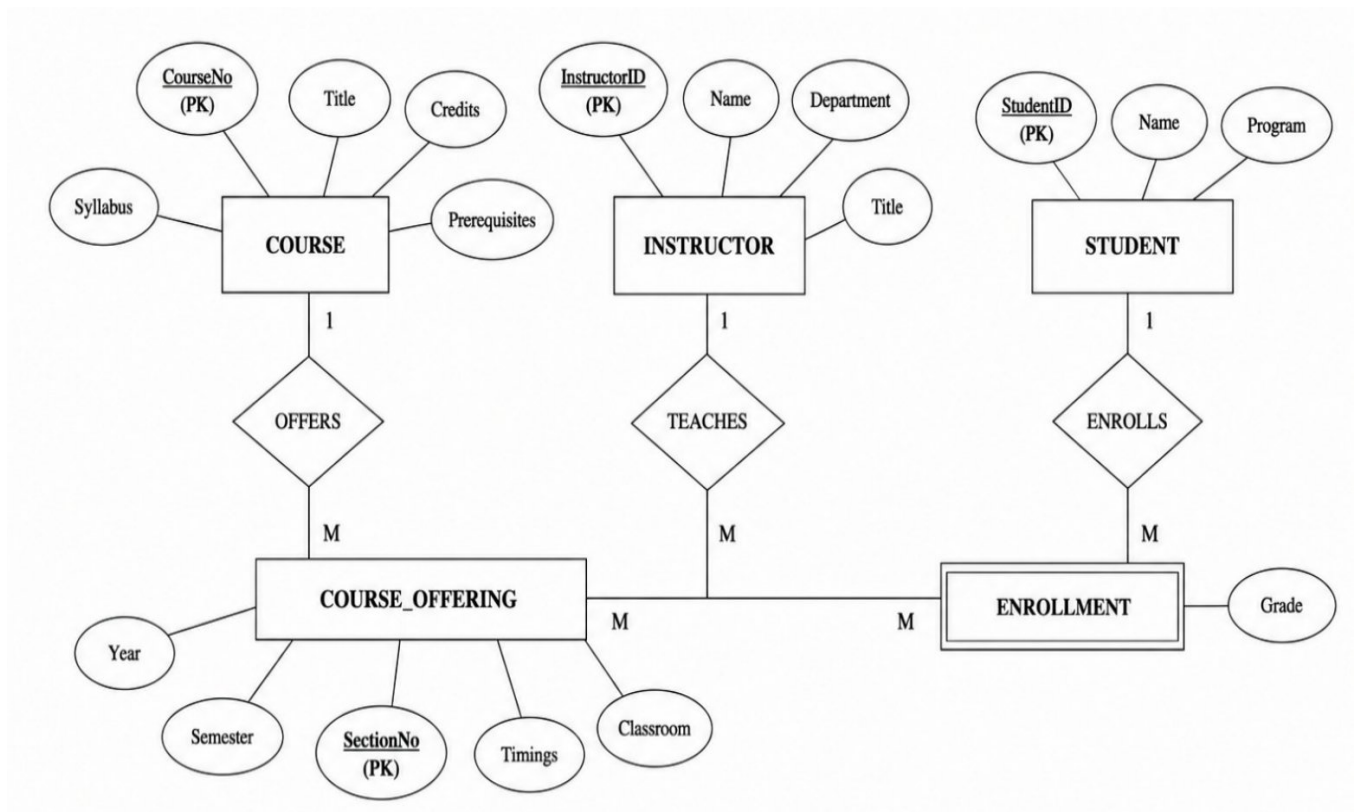
1. Ans:



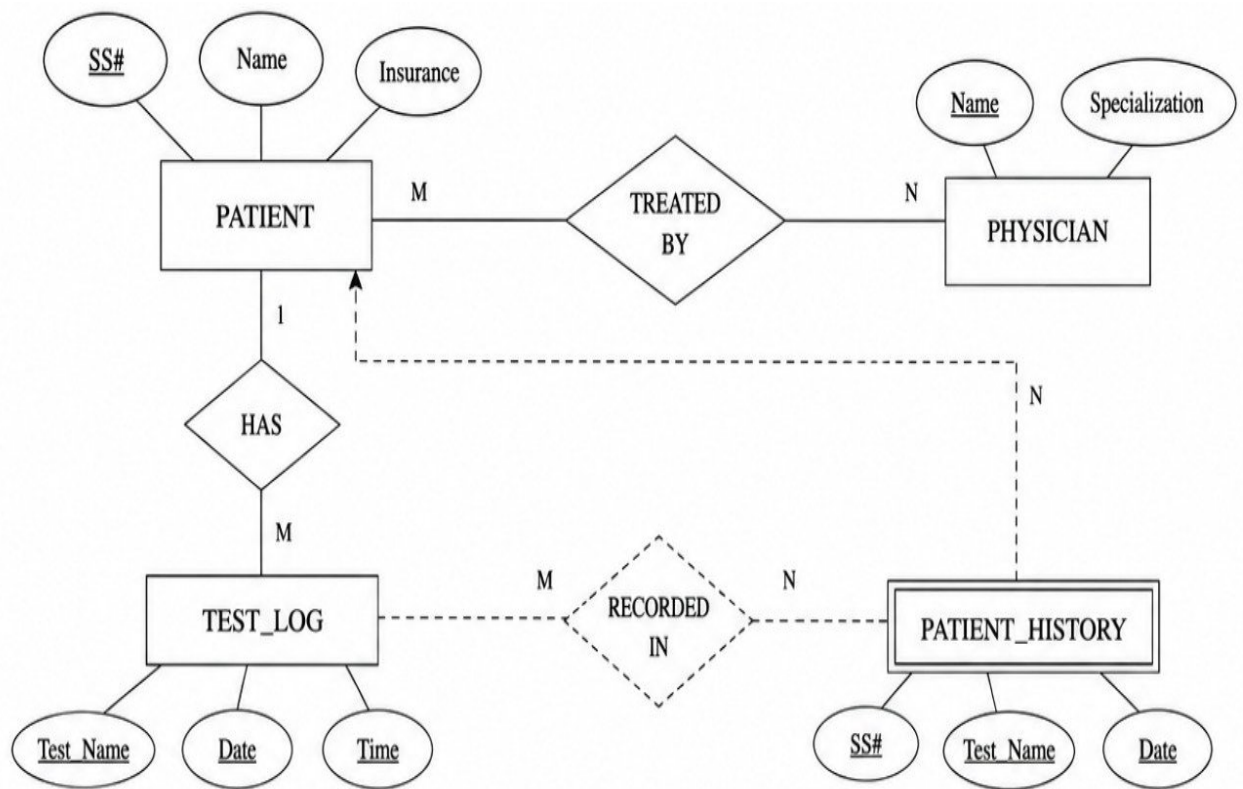
2 Ans:



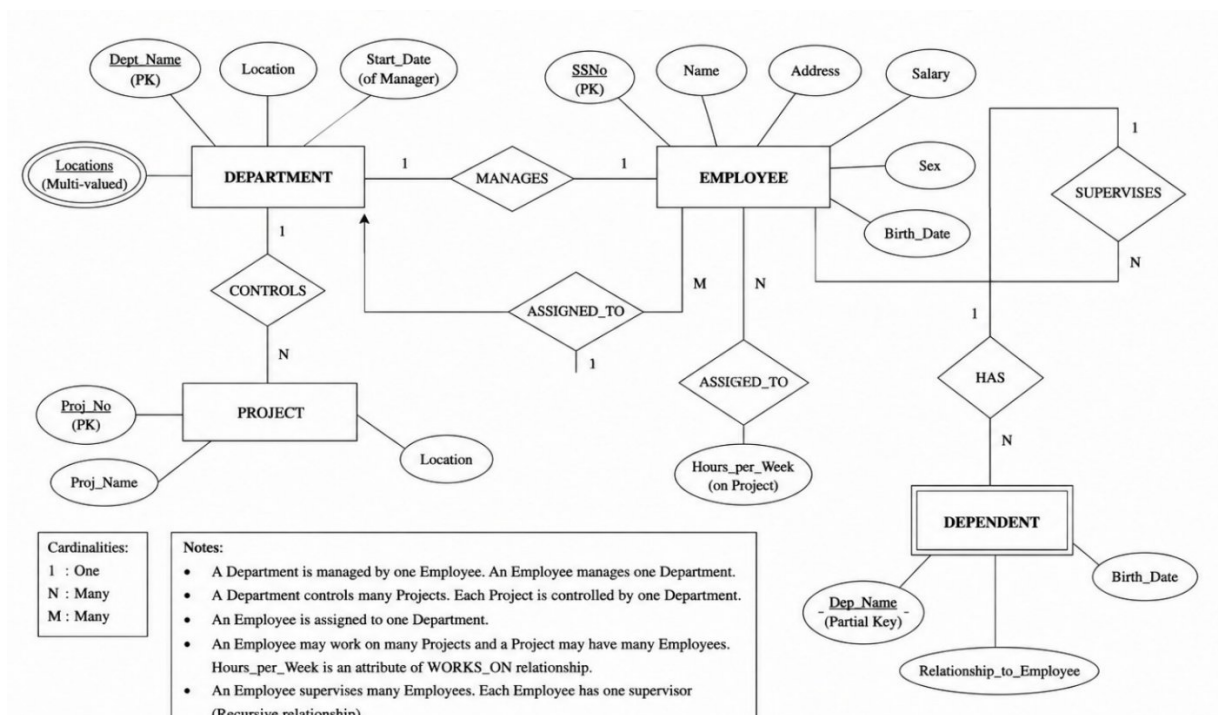
3. Ans:



4. Ans:



5. Ans:



6. Ans:

Schema

A schema is the blueprint or structure of a database. It defines tables, columns, and relationships.

Example:

Student(Student_ID, Name, Department, Age)

Instance

An **instance** is the actual data stored in the database at a particular time.

Example:

Student_ID	Name	Department	Age
101	Rahul	CSE	20

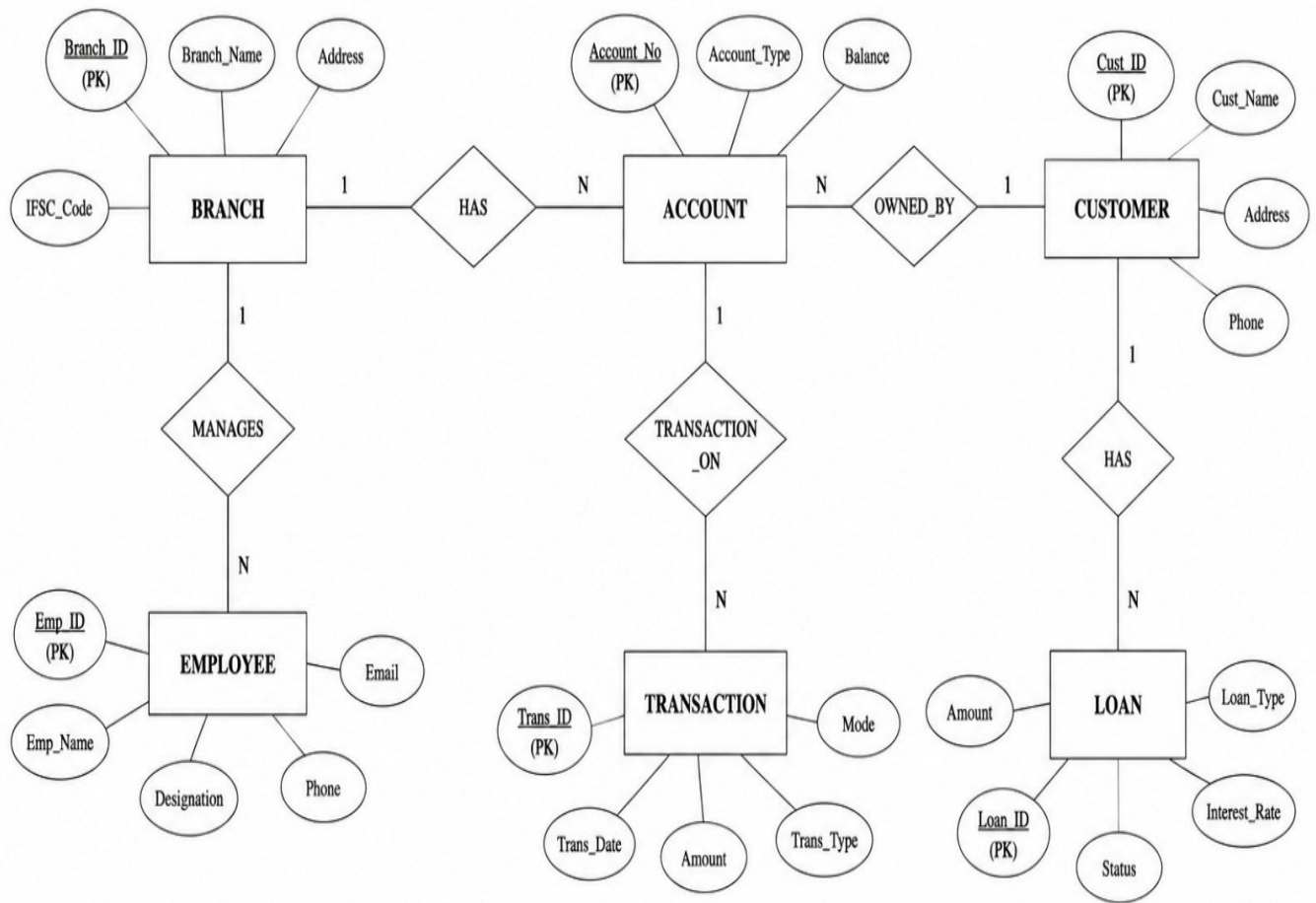
Schema vs Instance

Schema	Instance
Database structure	Actual data
Changes rarely	Changes frequently

Types of Schema

- **Logical Schema:** Defines tables, attributes, and relationships.
Example: Student(Student_ID, Name, Age)
- **Physical Schema:** Defines how data is stored on disk (files, indexes).
- **View Schema:** Defines what a user can see.
Example: Student view shows only Student_ID, Name, and Department (Age hidden).

7. Ans



8. Ans

Extended ER (EER) Model

The Extended Entity-Relationship (EER) model is an extension of the ER model. It includes advanced concepts like generalization, specialization, and aggregation to represent complex database relationships.

1. Generalization

- Generalization combines similar lower-level entities into a higher-level entity.
- It is a bottom-up approach.

Example:

Car and Bike → Vehicle

2. Specialization

- Specialization divides a higher-level entity into lower-level entities based on specific characteristics.
- It is a top-down approach.

Example:

Employee → Manager, Engineer, Clerk

3. Aggregation

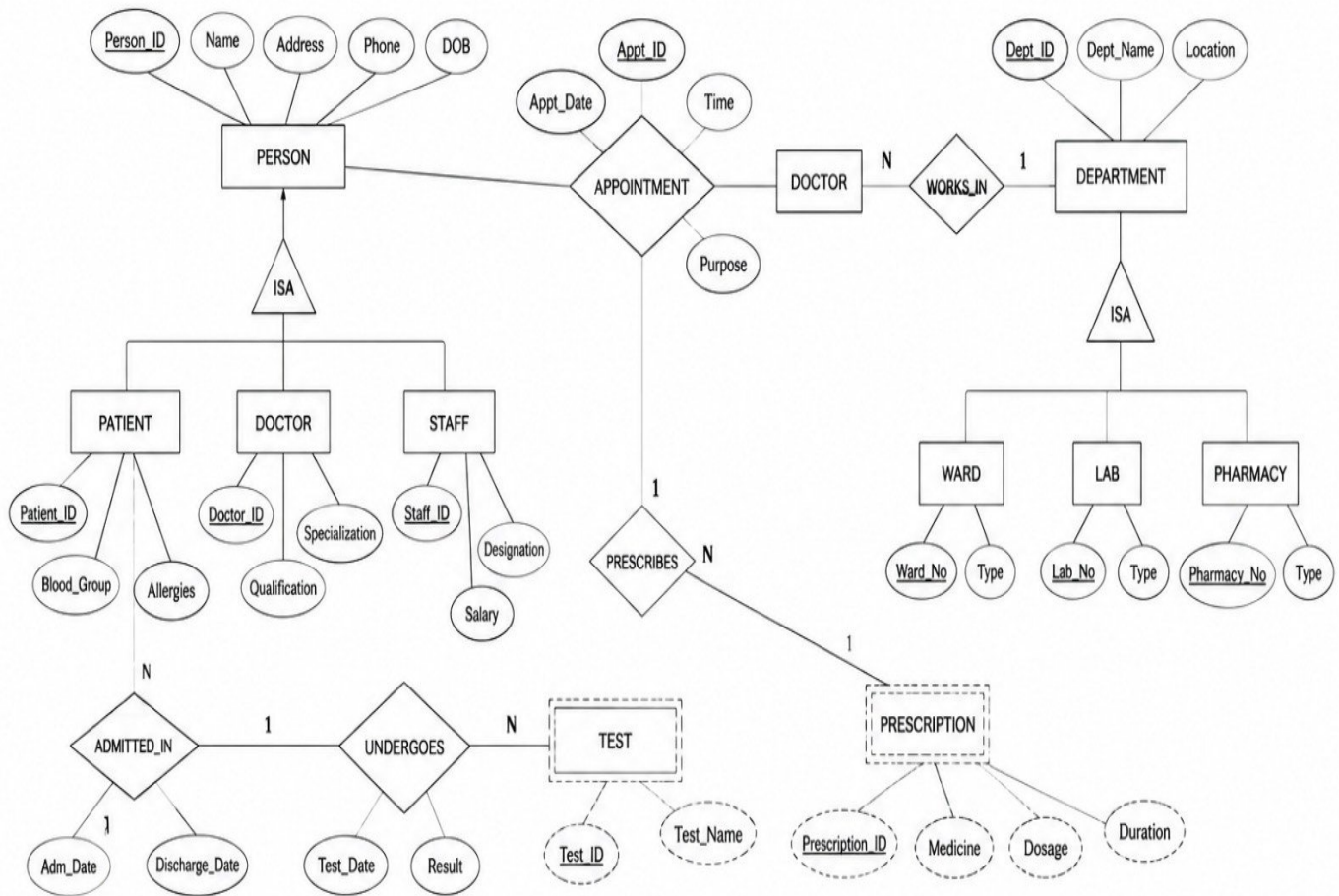
- Aggregation treats a relationship between entities as a higher-level entity so it can participate in another relationship.

Example:

An Employee works on a Project. This Works_On relationship is aggregated to relate with a Manager who supervises it.

Concept	Description	Example
Generalization	Combines entities into one higher-level entity	Car, Bike → Vehicle
Specialization	Divides an entity into sub-entities	Employee → Manager, Engineer
Aggregation	Treats a relationship as an entity	Manager supervises Employee–Project relationship

9. Ans:



10. Ans:

Role of a Database Administrator (DBA)

A Database Administrator (DBA) is responsible for managing, maintaining, and securing a database to ensure it operates efficiently and reliably.

Responsibilities of a DBA

- **Database Installation and Configuration:** Install and set up the DBMS.
- **Database Security:** Create user accounts and control access to data.
- **Backup and Recovery:** Perform regular backups and restore data when needed.
- **Performance Monitoring:** Optimize database performance and resolve issues.
- **Data Integrity:** Ensure data is accurate, consistent, and reliable.
- **Database Maintenance:** Update, monitor, and maintain the database regularly.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand